

**EXPERIMENT – 15 - EMBEDDED C PROGRAM TO CONVERT THE
HEXADECIMAL DATA 0XCFH TO DECIMAL AND DISPLAY THE
DIGITS ON PORTS P0, P1 AND P2**

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PROGRAM

```
#include <reg51.h>

void main(void)
{
    unsigned char hexa = 0x08;
    unsigned char hundreds, tens, units;
    ACC = hexa;
    B = 10;
    ACC = ACC / B;
    units = B;
    B = 10;
    ACC = ACC / B;
    tens = B;
    hundreds = ACC;
    P0 = units;
    P1 = tens;
    P2 = hundreds;
    while(1);
}
```

OUTPUT

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x07

Disassembly

```
9: units = B;
C:0x0806 AFF0 MOV R7,B(0xF0)
10: B = 10;
C:0x0808 75F00A MOV B(0xF0),#0x0A
11: ACC = ACC / B;
```

exp15.c

```
1 #include <reg51.h>
2 void main(void)
3 {
4     unsigned char hexa = 0xCF;
5     unsigned char hundreds, tens, units;
6     ACC = hexa;
7     B = 10;
8     ACC = ACC / B;
9     units = B;
10    B = 10;
11    ACC = ACC / B;
12    tens = B;
13    hundreds = ACC;
14    P0 = units;
15    P1 = tens;
16    P2 = hundreds;
17    while(1);
18 }
```

Parallel Port 0

Port 0: 7 Bits 0
P0: 0xFF

Parallel Port 1

Port 1: 7 Bits 0
P1: 0xFF

Parallel Port 2

Port 2: 7 Bits 0
P2: 0xFF

Command

Running with Code Size Limit: 2K
Load "C:\Keil_v5\CS1\Examples\HELLO\Objects\exp15"
WS 1, 'hexa
WS 1, 'hundreds
WS 1, 'tens
WS 1, 'units

Watch 1

Name	Value	Type
hexa	0x07	uchar
hundreds	0x00	uchar
tens	0x00	uchar
units	0x07	uchar

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

Simulation t1: 0.00039800 sec L10 C:1 CAP NUM SCRL OVR: R/W

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x07

Disassembly

```
16: P2 = hundreds;
C:0x0812 F5A0 MOV PPAGE_SFR(0xA0),A
17: while(1);
C:0x0814 80FE SJMP C:0814
133: MOV R0,#IDATALEN - 1
```

exp15.c

```
1 #include <reg51.h>
2 void main(void)
3 {
4     unsigned char hexa = 0xCF;
5     unsigned char hundreds, tens, units;
6     ACC = hexa;
7     B = 10;
8     ACC = ACC / B;
9     units = B;
10    B = 10;
11    ACC = ACC / B;
12    tens = B;
13    hundreds = ACC;
14    P0 = units;
15    P1 = tens;
16    P2 = hundreds;
17    while(1);
18 }
```

Parallel Port 0

Port 0: 7 Bits 0
P0: 0x07

Parallel Port 1

Port 1: 7 Bits 0
P1: 0x00

Parallel Port 2

Port 2: 7 Bits 0
P2: 0x02

Command

Running with Code Size Limit: 2K
Load "C:\Keil_v5\CS1\Examples\HELLO\Objects\exp15"
WS 1, 'hexa
WS 1, 'hundreds
WS 1, 'tens
WS 1, 'units

Watch 1

Name	Value	Type
hexa	0x07	uchar
hundreds	0x00	uchar
tens	0x00	uchar
units	0x07	uchar

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

Simulation t1: 0.00041100 sec L17 C:1 CAP NUM SCRL OVR: R/W